

CSP502 Software Lab-I - M.Tech. CSE W26
Lab Exercise 4B: Data Structures as Design Decisions
CHANGE REQUIREMENT

To be downloaded/opened only when released by the instructor during the practical.

IMPORTANT: Do not modify your program immediately. First complete Part A below and record your design response. Show/submit this response as instructed before making implementation changes.

Changed Requirement

Until now, your analyzer could load the complete log file and answer queries on the available data.

The system will now receive log records continuously. After every 1,000 newly received records, it must report the five users having the highest number of events up to that point.

You should not repeatedly process the complete log file from the beginning merely to produce this report.

Part A - Design Response Before Coding

Before modifying the implementation, answer the following based on your existing design:

1. Which assumption in your original data-structure decision has changed?

Response: _____

2. Which existing data structure(s) can still be retained?

Response: _____

3. What additional or alternative data structure, if any, is required?

Response: _____

4. What is the expected cost of processing a new event and producing the Top-5 report?

Response: _____

Record these decisions before implementation. Do not rewrite your original Practical 4B decision table; this response should show how your design thinking changes when the requirement changes.

Part B - Modify and Validate

After completing Part A and receiving permission/instruction to proceed:

- Modify your design and implementation to satisfy the changed requirement.

- Use the workload generated in Practical 4A and process the records as a continuing stream or sequence of incoming records.
- Produce the Top-5 report after every 1,000 newly processed records.
- Validate that the reported users and counts are correct.
- Record whether the observed behaviour agrees with the complexity/performance you expected in Part A.

Part C - Brief Post-Change Observation

- 1. What part of your original design could be reused without change?**
- 2. What part had to be redesigned, and why?**
- 3. Did the changed requirement justify introducing a more sophisticated data structure, or was a simpler approach sufficient for your actual workload? Support your answer using your workload and observations.**

Submission with Practical 4B

Include this change-response work with your Practical 4B submission: the pre-coding design response, modified implementation, validation evidence, and brief post-change observations.